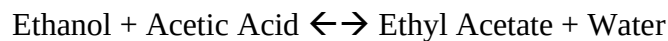


EXERCISE 4.1

Objective: Compare the use of different reactor types to model one reaction.

Problem Description

A stream containing 8.892 kmol/hr of water, 186.59 kmol/hr of ethanol and 192.6 kmol/hr of acetic acid at temperature of 70°C and pressure of 1 atm is fed to different types of reactors (RSTOIC, R-Gibbs, RPLUG and RCSTR). The reactor operating conditions, temperature is at 70°C and pressure at 1 atm. The stoichiometry and kinetic parameters are given below:



Kinetic Parameters:

- » Rate is in the unit of kmol/(m³.s)
- » Forward Reaction: Pre-exp. Factor = 1.9×10^8 , Act. Energy = 5.95×10^7 J/kmol
- » Reverse Reaction: Pre-exp. Factor = 5.0×10^7 , Act. Energy = 5.95×10^7 J/kmol
- » Reactions are first order with respect to each of the reactants in the reaction (second order overall).
- » Reactions occur in the liquid phase.
- » Composition basis is Molarity.
- » Hint: **Check that each reactor is considering Liquid as Valid phases.**

The process flowsheet with reactor specifications is shown in Figure 4.1 Simulate the process using NRTL-HOC thermodynamic model.

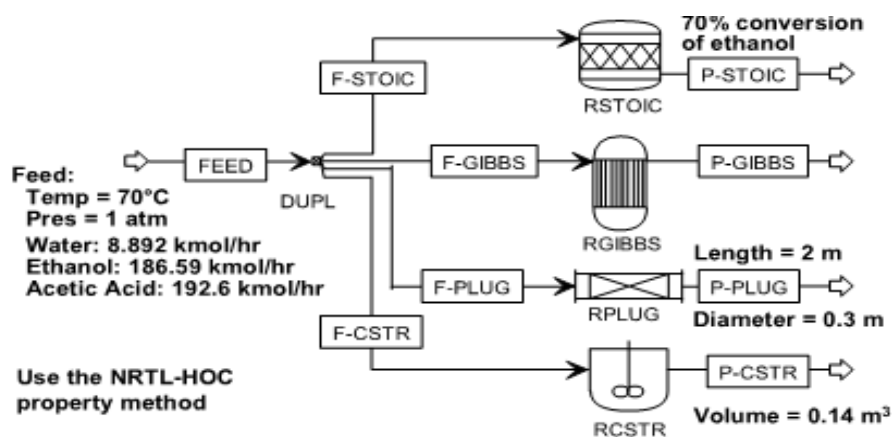
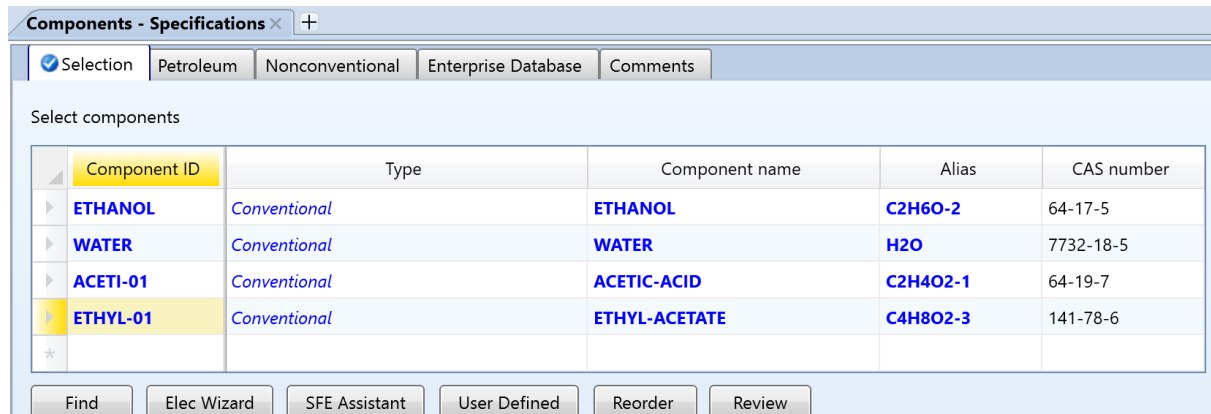
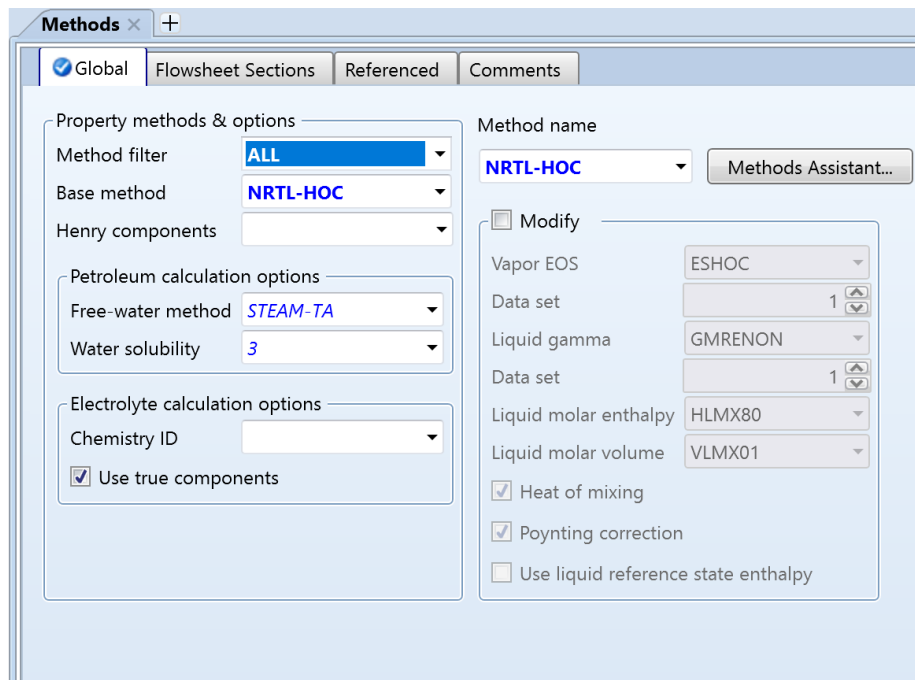


Figure 4.1

1. Create a new simulation using the General with Metric Units template.
2. In the Properties environment, add the components from the reaction. Add WATER, ETHANOL, ACETIC ACID AND ETHYL-ACETATE. From this point, ACETIC ACID will be referred to as AA and ETHYL-ACETATE as ETAC. You can use these names as their respective IDs.

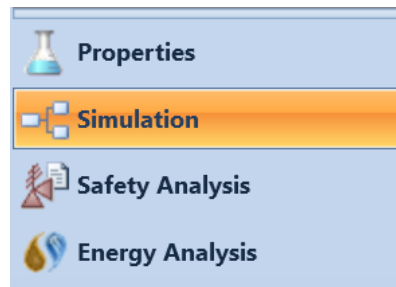


3. Select NRTL-HOC as the property method to be used in the simulation. NRTL-HOC is used to accurately represent the dimerization of acetic acid in the vapor phase.

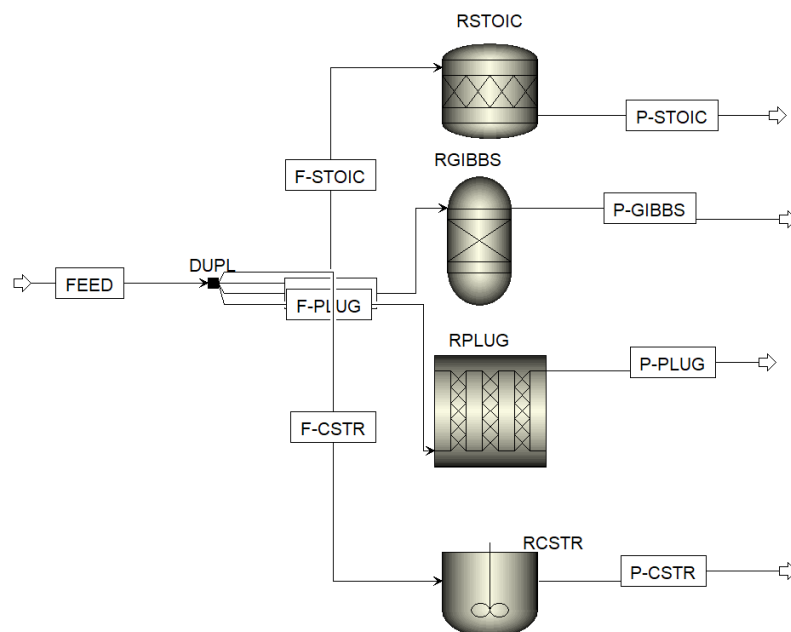


4. Click NEXT to run Property Analysis / Setup.

- Go to the simulation environment by clicking on the Simulation button below the navigation pane.



- To create the process flowsheet, add the RSTOIC, RGIBBS, RPLUG and RCSTR model located under the Reactors tab in the Model Palette, and name it accordingly (right-click on the stream and click rename or select the stream and press Ctrl+M). Add a material stream for the inlet and rename as FEED. Then, add a Stream Duplicator (DUPL) under Manipulators tab in the Model Palette. Connect the FEED stream to the inlet port of the duplicator. The Stream Duplicator can be used to duplicate the FEED stream and can be connected to different reactor blocks. Connect the inlet and outlet ports with Material streams and name them accordingly.



- Double-click the FEED stream to enter the input stream characteristics. Select Temperature and Pressure for Flash Type. Enter 1 atm for Pressure and 70 C for temperature. Change the Composition basis to Mole-Flow with unit kmol/hr. Enter the mole-flow 8.892 kmol/hr for water, 186.59 kmol/hr for ethanol and 192.6 kmol/hr for acetic acid.

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The screenshot shows the 'FEED (MATERIAL)' window in Aspen Plus. The 'Specifications' tab is active. Under 'Flash Type', 'Temperature' and 'Pressure' are selected. The 'State variables' section shows Temperature at 70 C, Pressure at 1 atm, Vapor fraction, Total flow basis as Mole, Total flow rate, and Solvent. The 'Reference Temperature' section has fields for Volume flow and Component concentration reference temperatures, both set to C. The 'Composition' section shows a table of components and their values in kmol/hr.

Component	Value
ETHANOL	186.59
WATER	8.892
ACETI-01	192.6
ETHYL-01	
Total	388.082

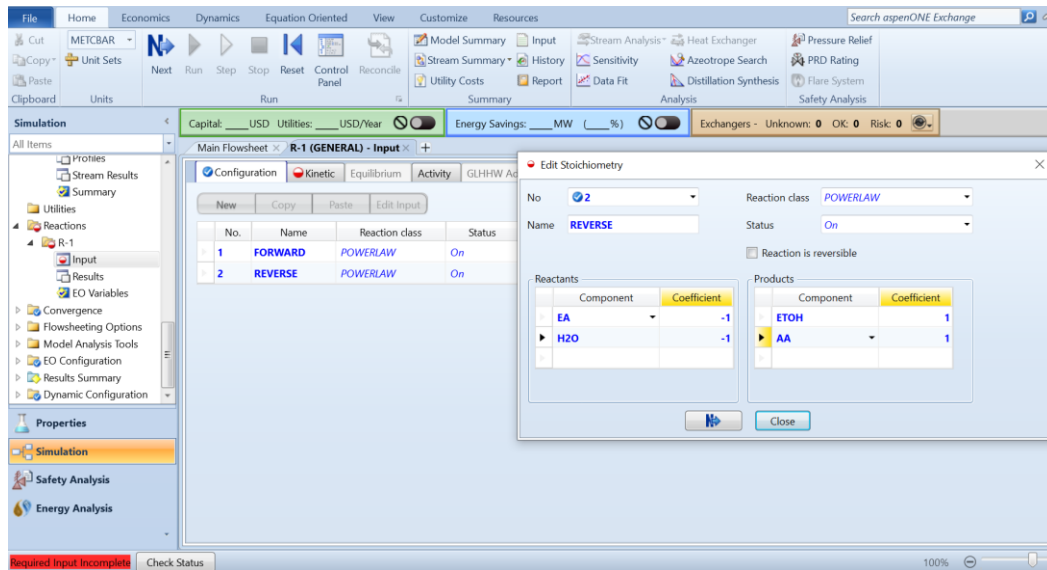
- Double-click on the reactor block to enter the specifications. Enter the reactors specification. Use the information given in the Figure 4.1.
- Include the reaction details. Create NEW reaction set R-1 by choosing GENERAL type of reaction. Then Create NEW reaction and include stoichiometry of forward reaction. Name the reaction as FORWARD.

The screenshot shows the 'Edit Stoichiometry' window for reaction R-1. The reaction is named 'FORWARD' and is classified as 'POWERLAW'. The reaction is set to be irreversible. The reactants are ETOH and AA, both with a coefficient of -1. The products are EA and H2O, both with a coefficient of 1.

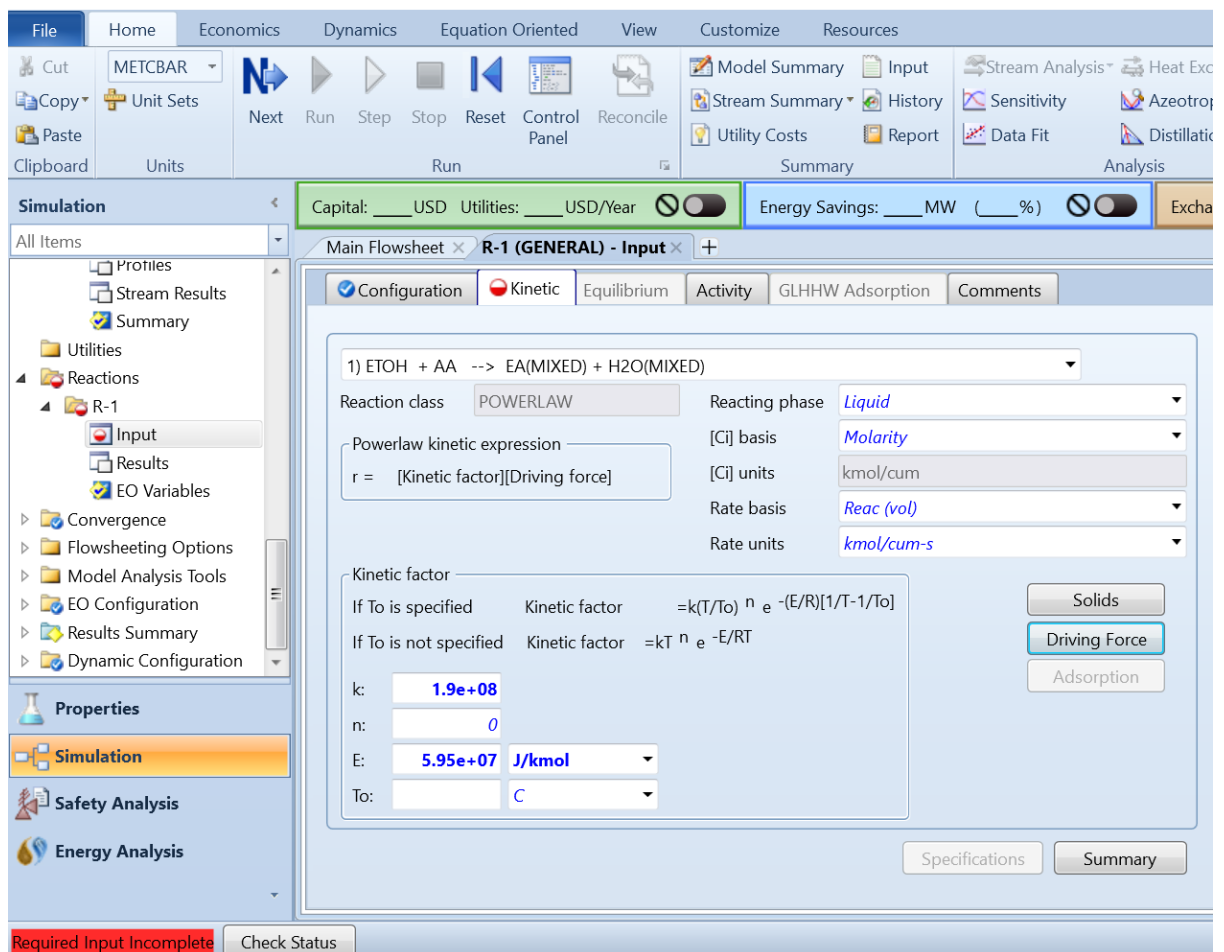
Reactants		Products	
Component	Coefficient	Component	Coefficient
ETOH	-1	EA	1
AA	-1	H2O	1

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10. Then, create new reaction for reverse reaction and include stoichiometry of reverse reaction. Name the reaction as REVERSE.



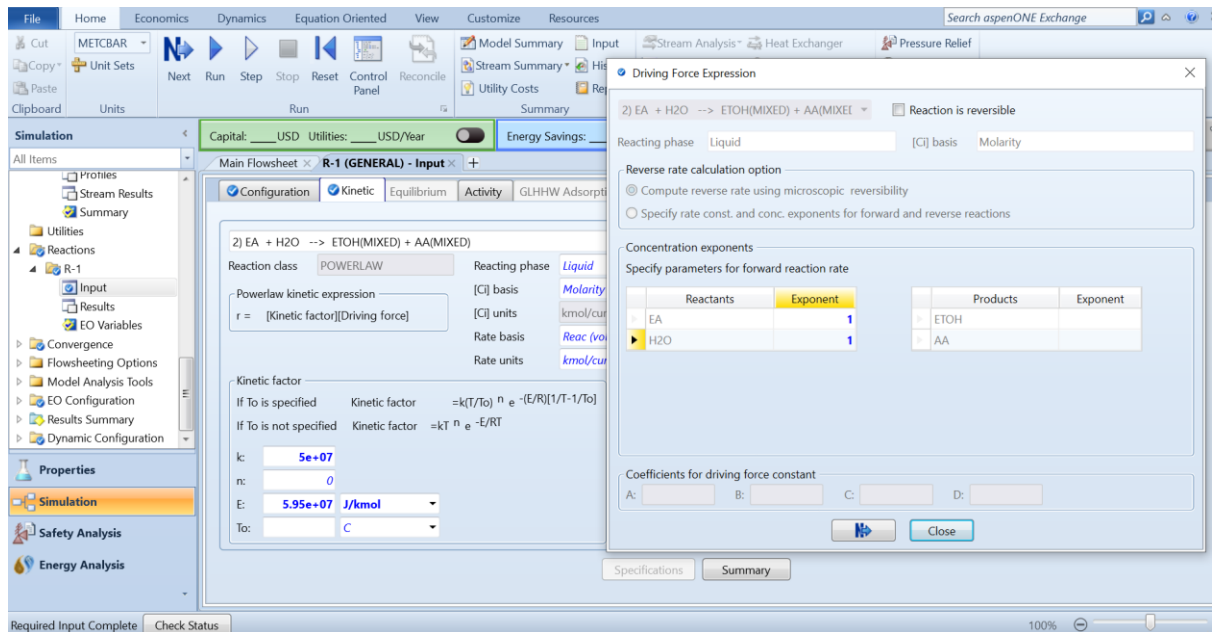
11. Include the kinetic and driving force of forward reaction. The kinetic and driving force can be specify under Reactions | Kinetic tab.



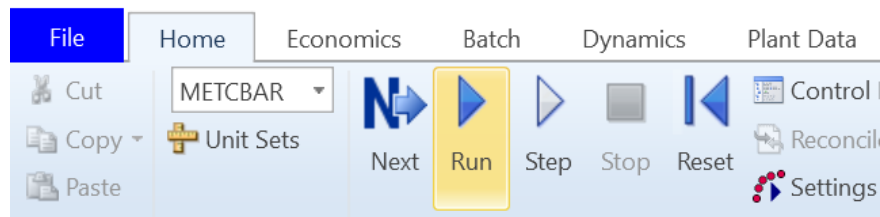
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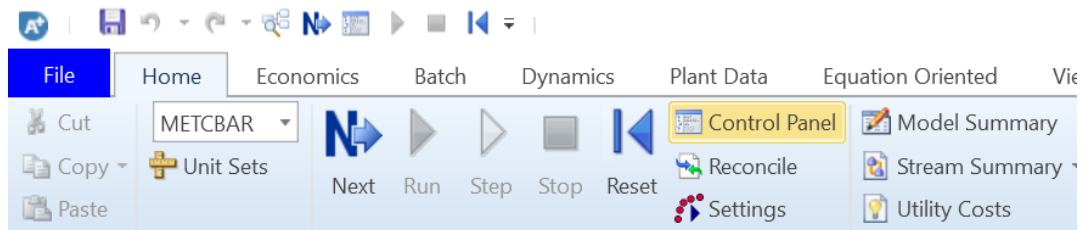
12. Include kinetic and driving force of reverse reaction.



13. Open the Control Panel and run the simulation by pressing the RUN button or press F5.



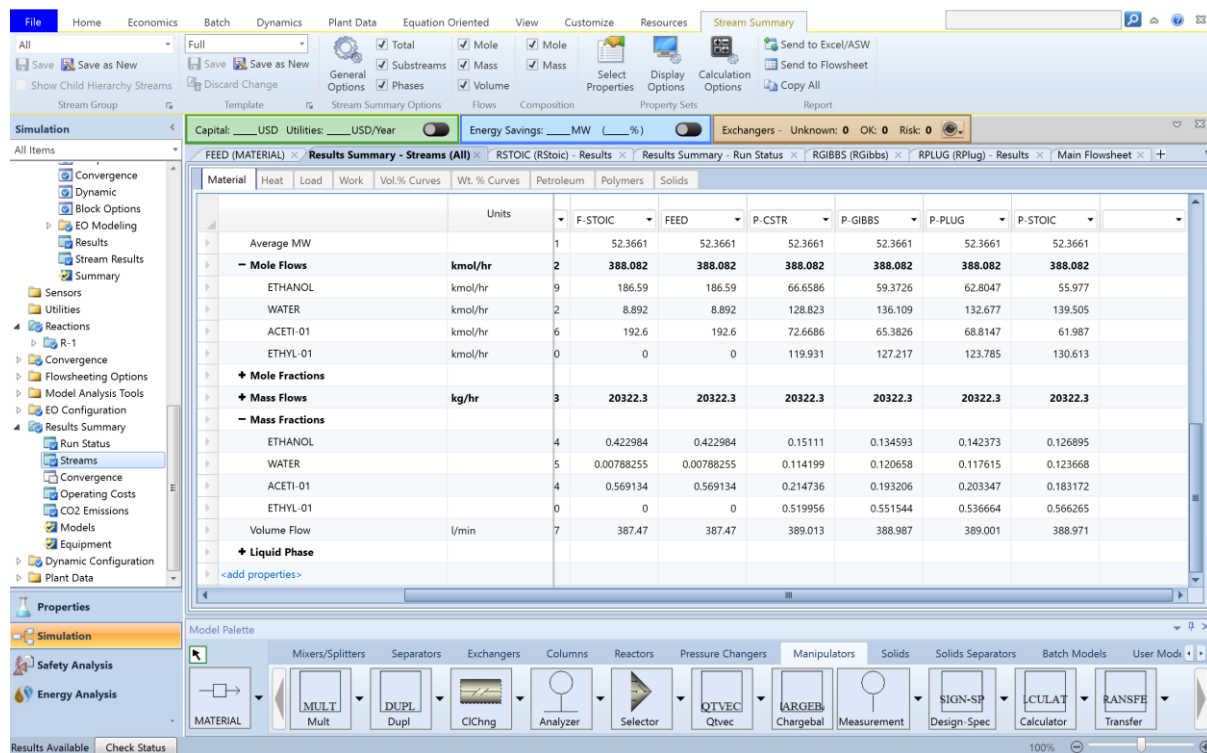
14. To view the simulation status, click on Control Panel in the Home navigation pane.



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15. Check the simulation results. To view the stream results, click on Result Summary | Streams.



16. Register the results:

	RStoic	RGibbs	RPlug	RCSTR
Amount of Ethyl Acetate produced (kmol/hr)				
Mass fraction Ethyl Acetate in product stream				
Heat duty (Gcal/hr)				

17. Save your file as EXERCISE4.1.BKP